

Kai Iniguez

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Profile

Electrical Engineering B.S. candidate graduating May 2027 with hands-on RF test, RF instrumentation, RF communication, antenna system design, and hardware integration experience. Skilled in VNA- and spectrum-analyzer-based characterization, automated test and report generation, and translating RF measurements into clear technical documentation

EDUCATION

California Polytechnic State University, San Luis Obispo

Bachelor of Science in Electrical Engineering

May 2027

3.5/4 GPA | Dean's List

RELEVANT COURSEWORK

Phased Array Antennas (Graduate), Wireless Communications, Communication Systems, Electromagnetic Waves, Classical Control Systems(& Lab), Semiconductor Device Electronics(& Lab), Digital Electronics & Integrated Circuits(& Lab)

SKILLS

- **Hardware:** VNA (Network Analyzer), Spectrum Analyzer, Oscilloscope, Source Meter, Multimeter, RF Signal Generator, PCB Design
- **Applications:** Ansys HFSS, Keysight ADS, LTSpice, Eagle, Fusion360, Microsoft Excel
- **Software:** Python, MATLAB, C programming, SystemVerilog, Arduino

EXPERIENCE

EMC Test Engineer Intern, Hewlett Packard Enterprise

June 2026-August 2026

- Characterized CDNs and LISNs with a VNA, measuring insertion loss and impedance across 150kHz-30MHz
- Characterized EFT/burst generator with an oscilloscope, verifying pulse and impulse parameter requirements
- Developed Python test automation using PyVISA and SCPI to control VNAs and oscilloscopes, reducing test and report time by 75%.
- Supported RF/EMC characterization using spectrum analyzers and 10-meter anechoic-chamber measurements

TECHNICAL PROJECTS

28 GHz 5G mmWave Phased-Array Beamforming and Beam-Steering System

August 2026-Present

- Designing a 16-element phased-array model for 28 GHz 5G mmWave beamforming applications
- Simulating electronic beam steering and evaluating gain, beamwidth, sidelobes, and scan loss using Ansys HFSS
- Assessing phase-shifter errors and failed antenna elements to characterize beamforming resilience
- Relating array-level tradeoffs to practical 5G base-station and handset design constraints

Automatically Tuned HF Magnetic Loop Antenna | Senior Project Design

August 2026-Present

- Designing and optimizing a compact magnetic loop antenna for 7-14 MHz HF operation for Hascall Denke's military use
- Developing a nominal 50-ohm matching network and electrically actuated resonance-tuning system
- Simulating antenna geometry and RF performance in Ansys HFSS, including gain, bandwidth, efficiency, and radiation pattern
- Collaborating with a three-person team to prototype, test, and build the antenna system

RC Car Embedded Control System

January 2026-June 2026

- Developed embedded firmware in C for a STM32 microcontroller RC car, integrating motor control, user input and peripheral interfaces
- Implemented real-time control using GPIO, PWM, timers, interrupts, and serial communication for responsive vehicle operation
- Integrated and debugged wireless-control, motor-driver, and steering hardware to improve system reliability

OTTER CPU and Assembly Programming Project

January 2025-June 2025

- Designed a fully functional custom CPU in SystemVerilog, using a modular datapath, control unit, ALU, and instruction-execution logic
- Developed a custom ISA and assembly language; integrated VGA and keypad I/O to deploy a Whack-A-Mole game

INVOLVEMENT

High Voltage Subteam Member, Cal Poly Racing

September 2025-June 2026

- Designed and fabricated portions of the Tractive System Active Light (TSAL) PCB using Altium
- Contributed to high-voltage wiring, safety systems, and subsystem integration on a Formula SAE electric vehicle

Member, Power Engineering Society

January 2024-Present